MSCA Digitial Finance

"Multi-criteria decision making in sustainable finance", Kaiserslautern, 26.-28.10.2026

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Lecture 3: Portfolio Optimization with Sustainable Assets

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Aims of this Lecture:

- Introduction to special aspects of portfolio optimization with sustainable assets and sustainability requirement
- Showing you concepts how the state could steer investments in a sustainable direction
- Presenting a framework that has a feedback to climate scenarios and is related to an asset allocation strategy for insurers.
- Complaining a bit on legal restrictions and on green washing tendencies, …

Optimal investment with climate scenarios – Motivation

Why should life insurers invest in green assets?

Because they can …

- Huge amounts of money available for investment with a long duration
- Pension is foreward looking („Spend your retirement in an intact environment“)
- Life insurers already own solar parks, windparks, …
- Alternative investments are good for diversification
- Life insurers are accurately watched for their bevaviour with regard to sustainability

… and life insurers can make their customers aware of the fact that they can actively help to contribute against the climate catastrophe!

Optimal investment with climate scenarios – Sustainability ratings

Bloomberg Rating.

ESG-scorings between 0 und 100 (100 = best value) for 11,500 companies from 83 countries based on approx. 800 criteria (see Heinke (2021))

Heinke (2021): Extends the Bloomberg rating to further institutions, countries and companies that are not covered by the Bloomberg rating

But:

How about investment in local infrastructure? How about recycling? How about improvement of air and water quality? …

Is the whole rating business not flexible enough and far to expensive?

Here:

We assume that there is a sustainability rating. Legal and/or political aspects are beyond our competence … **but we still comment on them in the next slides.**

Why should one invest in sustainable assets, why in companies that are treating their workers in a fair way and that have a fair and legal governance?

Because their is no long-term alternative

- for **us**
- for **our children**
- for **our environment**
- for **our safety** and for **our good conscience**

Also:

- There should be a **high correlation** between **insurance** business and **climate** evolution (property and casual insurance, health insurance, life insurance, …)
  => Use this for **risk management** and **investment opportunities**
- The relation of insurance companies‘ investments in climate related issues gives them (and their products!) a **modern, innovative and positive image** as **partner** and as **employer**

What are (legal) problems of sustainable products and projects?

Biggest problems

- It is often legally(!) impossible to advertise insurance products as „sustainable products“ (Reasons are complex, hard to prove (!) that problems are in line with the EU taxonomy, …)
- There are no official certificates for green products
- There is also a big sensitivity for „green washing“

Possible solutions

Use neutral names for funds/products, but describe them in a detailed/proud way as s

- We **invest** your premia **according to strict ESG criteria of exclusion** (i.e. no weapons, no fossile energy, …)
- Insurer gives an **annual report** on which **ecological projects** have been **financed with the contributions** (expansion of renewable energy plants, afforestation, …)
- Contracts contain „**sustainability clauses**“ (i.e. bonus for sustainable replacements, …)

Possibly more good ideas to act well and talk about them!

Some hope for clarity …

Institution for standards: International Sustainability Standards Board (ISSB) (since 2021)

Aims: setting accounting standards for sustainable activities

- develops global minimum standards that have to be reported (IFRS S1, IFRS S3)
- IFRS S1 is concerned with financial risks and chances of the company that result from environment, social and governance activities/style.
- IFRS S2 requires a detailed quantitative report on climate related actions and measures (direct/indirect CO^2 emissions), physical climate risks (storm, flood, fire, …), scenario analyses for the company's strategy, transition plans, etc …
- …

Non-representative aspects from the recent literature, empirical research and tendencies on the Swiss Insurance market

- Scientific literature has switched from a somewhat ideologic view to a neutral view that focuses on return/risk, regulation, …
- Common view of market and science: sustainable investment is an accepted and established topic
- Sustainable assets do not necessarily perform worse than conventional assets and they allow to hedge risks that conventional assets cannot hedge
- Reporting efforts are increasing (see above)
- Positive steps: https://svv.ch/en/self-regulation-greenwashing
- List of companies https://svv.ch/sites/default/files/media/documents/2026-04/20260430_Register_unterstellten_Versicherungsgesellschaften_Selbstregulierung_Greenwashing_0.pdf
- https://svv.ch/sites/default/files/media/documents/2025-05/Greenwashing_Selbstregulierung_DE.pdf
- interesting: Stüttgen/Mattmann (2026) IFZ Sustainable Investments Studie 2025

Optimal investment with climate scenarios – Possible problems to solve

Aim: Setup concepts for

- Portfolio problems including **sustainability constraints**:
  Korn R., Nurkanovic A. (2023) Optimal portfolios with sustainable assets: Aspects for life insurers. European Actuarial Journal **13**, 125–145.
- Motivate a **taxation/subsidy-concept** such that the investors automatically invest in sustainable assets
  Korn R., Nurkanovic A. (2025) Optimal portfolios and sustainable taxation. European Actuarial Journal (to appear). https://doi.org/10.1007/s13385-025-00423-w
- Develop a **realistic framework** where the insurance industry invests to **cope with the climate development**
  Korn R. (2025) Optimal portfolios for life insurers and climate scenarios. European Actuarial Journal. European Actuarial Journal **15**, 1-13.

What are climate scenarios? (from: SAA Sustainability Working Group (2024))

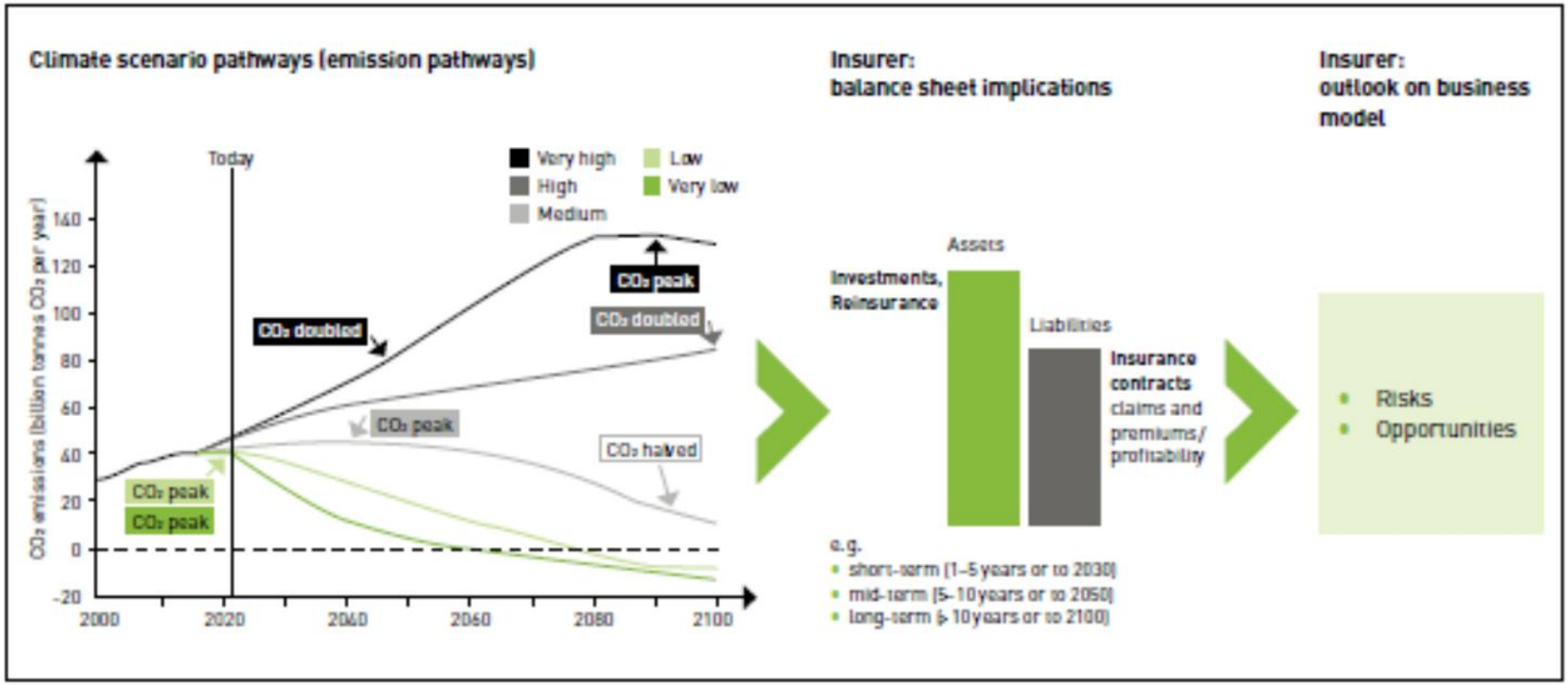


Figure 1

Source: left picture: Climate Science: A Summary for Actuaries, IAA; March 2022; other pictures: own design

Sustainable assets and life insurers – A simple portfolio problem

Black-Scholes-type market with money market account, *n* risky assets, prices given by

$$dB(t) = rB(t)dt, \;\; dS_i(t) = S_i(t)\left[b_i dt + \sum_{j=1}^{i} \sigma_{ij} dW_j(t)\right]$$

Asset *i* with sustainability rating $R_i(t) \in [0,1]$ => sustainability rating of portfolio π:

$$R(t) := R^{\pi}(t) = (1 - \pi(t)'\mathbf{1})R_0 + \sum_{i=1}^{n} \pi_i(t) R_i(t)$$

Let D(t) be the (portfolio-) **demand** at time t for sustainable investment

$\Rightarrow$ Basic portfolio problem with sustainability constraint in continuous time:

$$max_{\pi(.) \in A(x)} E_{0,x}\left(ln\left(X^{\pi}(T)\right)\right) \;\; s.t. \; R(t) \geq D(t) \;\; \forall t \in [0,T]$$

$$dX^{\pi}(t) = X^{\pi}(t)\left(\left(r + \pi(t)'(b - r1)\right)dt + \pi(t)'\sigma dW(t)\right)$$

Sustainable assets and life insurers – Solution of the basic problem

Proposition 1: Under the assumption of the max-offer condition

$$D(t) \leq \max\{R_0, R_1(t), \ldots, R_n(t)\}$$

the optimal portfolio process equals

$$\pi_S^{opt}(t) = \begin{cases} (\sigma\sigma')^{-1}(b - r\mathbf{1}), & if\ we\ have\ \ R_0 + (R(t) - R_0\mathbf{1})'(\sigma\sigma')^{-1}(b - r\mathbf{1}) \geq D(t) \\ (\sigma\sigma')^{-1}\left[(b - r\mathbf{1}) + \dfrac{D(t) - R_0 - (R(t) - R_0\mathbf{1})'(\sigma\sigma')^{-1}(b - r\mathbf{1})}{(R(t) - R_0\mathbf{1})'(\sigma\sigma')^{-1}(R(t) - R_0\mathbf{1})}(R(t) - R_0\mathbf{1})\right], & else \end{cases}$$

Note: In the second case and independent stocks, we see that their positions will only be reduced if they have a rating smaller than the one of the money market account/ actuarial reserve fund (ARF). They will only be **increased** if they have **a sustainability rating exceeding the one of the ARF**.

Sustainable assets and life insurers – Some numbers

Ex. 1: d=2, independent stocks, sustainability rating and demand fixed, $\mathbf{R_0=0}$

If the unconstrained optimal portfolio π^* does not satisfy the sustainability constraints, then the optimal portfolio under the constraint is given by

$$\pi_1^* \to \frac{b_1 - r + cR_1}{\sigma_1^2}, \pi_2^* \to \frac{b_2 - r + cR_2}{\sigma_2^2}, \quad c = \frac{D - \pi_1^* R_1 - \pi_2^* R_2}{\frac{R_1^2}{\sigma_1^2} + \frac{R_2^2}{\sigma_2^2}}$$

For D=0.2, r=0.01, b_1=0.03, b_2 = 0.04, σ_1=0.2 , σ_2= 0.3, R_1=0.1 , R_2=0.3 we obtain

$$\pi_1^* = 0.5 \to \frac{0.02 + 0.004}{0.04} = \mathbf{0.6}, \pi_2^* = \frac{1}{3} \to \frac{0.03 + 0.012}{0.09} = \mathbf{0.467}$$

Hence, **the (exp.) portfolio return increases** from 0.03 to 0.036, while the portfolio volatility also increases from 0.14 to 0.18. If we additionally require a non-negative position in the money market account, then the optimal sustainable portfolio equals (0.5, 0.5).

Sustainable assets and life insurers – Some numbers

Ex. 2: d=2, non-sustainable stocks, sustainable ARF (rest as in Ex. 1)

If the unconstrained optimal portfolio π^* does not satisfy the sustainability constraints, then the optimal portfolio under the constraint is given by

$$\pi_1^* \to \frac{b_1 - r - cR_0}{\sigma_1^2}, \pi_2^* \to \frac{b_2 - r - cR_0}{\sigma_2^2}, \quad c = \frac{D - R_0(1 - \pi_1^* - \pi_2^*)}{R_0^2} \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

The choices of D=0.2, r=0.01, b_1=0.03, b_2 = 0.04, σ_1=0.2 , σ_2= 0.3, R_0=0.3, R_1=0 , R_2=0 yield

$$\pi_1^* = 0.5 \to \mathbf{0.154}, \pi_2^* = \frac{1}{3} \to \mathbf{0.179}, \quad \pi_0^* = \frac{\mathbf{2}}{\mathbf{3}}$$

Consequence:

Portfolio return decreases from 0.03 to 0.018 as does the **portfolio volatility** from 0.14 to 0.062.

Sustainable assets and life insurers – More general utility functions

Now consider the class of power utility functions given by $U(x) = \frac{1}{\gamma}x^{\gamma}$, $1 > \gamma \neq 0$.

Then, we obtain:

Proposition 2: Under the assumption of the max-offer condition, constant demand for sustainability, constant sustainability rating for the assets and

(+) we either have $R^{\pi^*}(t) \geq D \ \forall t \in [0,T]$ or $R^{\pi^*}(t) < D \ \forall t \in [0,T]$.

Then, the optimal portfolio process equals

$$\pi_S^{opt}(t) = \frac{1}{1-\gamma}$$

$$= \begin{cases} (\sigma\sigma')^{-1}(b - r\mathbf{1}), & if\ we\ have\ \ R_0 + (R(t) - R_0\mathbf{1})'(\sigma\sigma')^{-1}(b - r\mathbf{1}) \geq D(t) \\ (\sigma\sigma')^{-1}\left[(b - r\mathbf{1}) + \frac{(1-\gamma)(D(t) - R_0 - (R(t) - R_0\mathbf{1})')(\sigma\sigma')^{-1}(b - r\mathbf{1})}{(R(t) - R_0\mathbf{1})'(\sigma\sigma')^{-1}(R(t) - R_0\mathbf{1})}(R(t) - R_0\mathbf{1})\right], & else \end{cases}$$

Voluntary optimal sustainability – Steering via subsidies/taxation by the state?

How can we obtain that the optimal unconstrained portfolio already satisfies the sustainability requirement?

By having suitable market coefficients of the different assets …

Idea:

Drift reduction (increase) via (de-) taxation of dividends of non-sustainable assets“

- Reduction/increase should ensure that the unconstrained optimal portfolio satisfies the sustainability constraint
- **Step 1:** Determine the necessary reductions/increases
- **Step 2:** How to bring Step 1 into reality, i.e. how should the government come up with a simple requirement?

=> Step 1 is easy to do, Step 2 does not seem to be so easy…

Suggestion for Step 2: Give a global lower bound D for the portfolio sustainability rating?

Voluntary optimal sustainability – Theoretical result

Proposition 3: Under the assumptions of Proposition 1 assume that the unconstrained optimal portfolio violates the sustainability requirement. Then, there exists a Girsanov transformation such that with the change of drift from *b* to *b**(t) with

$$b^*(t) = b + \frac{D(t) - R_0 - (R(t) - R_0\mathbf{1})(\sigma\sigma')^{-1}(b - r\mathbf{1})}{(R(t) - R_0\mathbf{1})(\sigma\sigma')^{-1}(R(t) - R_0\mathbf{1})}\,(R(t) - R_0\mathbf{1})$$

the optimal portfolio proess $\pi^*(t)$ in the transformed market given as

$$\pi^*(t) = (\sigma\sigma')^{-1}(b^*(t) - r\mathbf{1}),$$

already satisfies the sustainability constraint.

Interpretation: Change of drift can be seen as „(De-)taxation“ (with sign) of dividends of companies according to their sustainability rating.

Voluntary optimal sustainability – Again, Example 2

How can we obtain that the optimal unconstrained portfolio already satisfies the sustainability requirement?

By having suitable market coefficients of the different assets …

Idea:

Drift reduction (increase) via (de-) taxation of dividends of non-sustainable assets“

D=0.2, r=0.01, b_1=0.03, b_2 = 0.04, σ_1=0.2 , σ_2= 0.3, R_0=0.3, R_1=0 , R_2=0 will lead to

$$b_1 \to b_1 - r - cR_0 = \mathbf{0.00536},\ b_2 \to b_2 - r - cR_0 = \mathbf{0.01536},\ c = 0.0732$$

Note:

- Reduction/Increase is **independent** of the **volatility** of the assets, but has different effects on the portfolio components
- In the case of sustainable assets the following can happen
  - Short selling of a „weakly sustainable“ asset
  - Increase of the drift of a „strongly sustainable“ asset

Why have we considered this seemingly strange problem of voluntary optimal sustainability?

- It will help us to show how the state could possible implement behaviour of the investors via a suitable conditional law
- It also gives a hint how to steer the market in a suitable direction

Note: („Taxation/subisidy vs. risk/return effects“**)**

As not all market participants have the same utility function as the state (characterized by its given taxation/subsidy strategy), they will still solve **their own portfolio problem** (with a possibly different γ) and will deviate from the optimally taxed/subsidized position.

Optimal portfolios with climate scenarios

What are climate scenarios? (from: SAA Sustainability Working Group (2024))

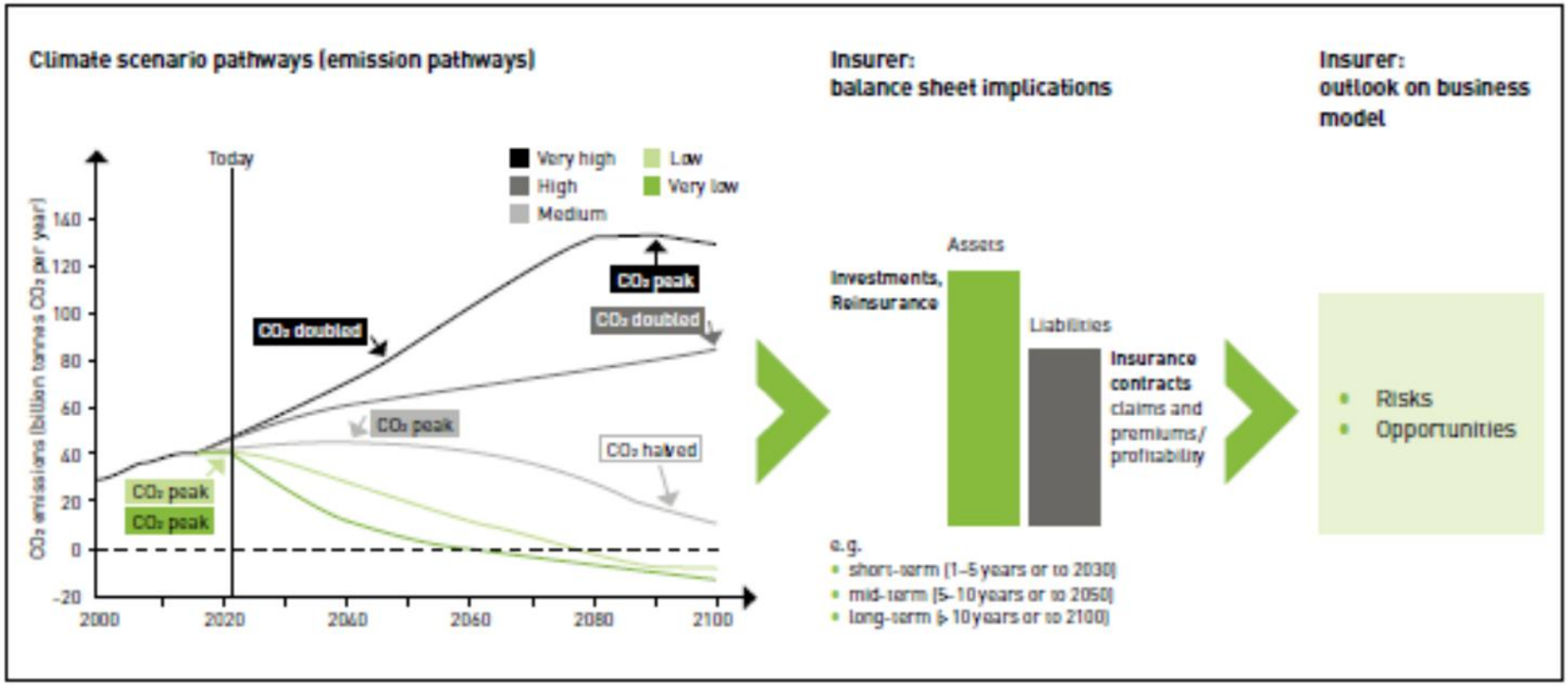


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Optimal portfolios with climate scenarios – Consequences?

Which consequences are implied by the realization of different climate scenarios?

- Governments will take **consequences** **in dependence of the realized climate scenario**
- **Here:** Possible **climate-dep. Taxation/subsidies** of the companies‘ gains (!) with regard to their contribution to global warming
- **Ass.:** The consequences at time T depending on the realized climate scenarios **are already known today**
- **Possible modelling approach:** Stock prices with climate depending drift parameters

$$dS_i(t) = S_i(t)\left(b_i(t, K(t))dt + \sum_{j=1}^{i} \sigma_{ij} dW_j(t)\right)$$

where *K*(t) is an indicator of the state of the climate as e.g. expressed in an average increase of the (global or local) temperature.

=> Why is this relevant for a life-insurer?

Optimal portfolios with climate scenarios – Climate-depending drift parameter

Advantage of the model framework:

Maximization of the Log-utiltiy works pathwise, again. Hence:

$$\pi^*(t) = (\sigma\sigma')^{-1}(b(t, K(t)) - r1)$$

is the optimal unconstrained portfolio process. The constrained one can be obtained „as above“.

Question: **But how do we get the constraints? Are there still any constraints?**

Note:

- Life insurers are very **large investors**. They cannot perform big reallocations without **large losses** !
- A forward looking abandoning of **wrong yields** can be beneficial!

Optimal portfolios with climate scenarios – Climate-depending drift parameter

Explicit modelling of the climate-depending drift parameter:

Idea:

As the temperature is essentially observable and thus also the government decision at T, the drift parameter should be a continuous function in the temperature (and time t).

Obvious choice for the drift functions:

$$b\big(t, K(t)\big) = E(b(T, K(T))|F_t) = \sum_{j=1}^{m} b^{(j)} P(Scenario\ j|F_t),$$

where $b^{(j)}$ are the final drift parameters such that the unconstrained optimal portfolio processes in scenario j are admissible with respect to the announced government decision.

=>

We have to estimate the probabilities of occurrence of the scenarios. This, indeed, depends on the explicit modelling of the temperature dynamics $K(t)$.

Optimal portfolios with climate scenarios – A path of a simplied example

Example: d=1, r = 0.02, b = 0.05, σ = 0.3, R_0 = 0.1, R_1 = 0.4, T = 6 (i.e. 2032)

=> π^* = 1/3 => R* = 0.2

Scenarios

Global warming 2026: K(0) = 1°C

S_1: D(T) = 0.3, if K(6) ≥ 1.5°C => π_1 = 2/3, i.e. b_1 = 0.08

S_0: D(T) = 0.2, if K(6) < 1.5°C => π_0 = 1/3, i.e. b_0 = 0.05

Climate dynamics/parameter:

$$\mathrm{dK(t)} = \mathrm{K(t)}[0.1\,dt + 0.1\,dW^*(t)],\;\; K(0) = 1$$

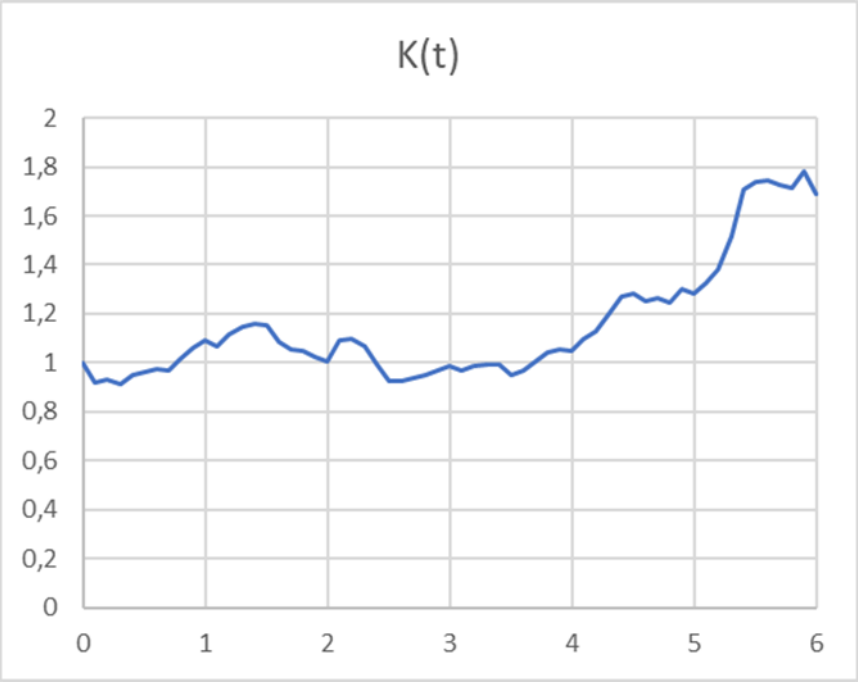
$$\Rightarrow \mathrm{E\big(K(t)\big)} = e^{0.1t},\;\; \mathrm{P}(K(T) \geq 1.5| \geq K(t)) = \Phi\left(\frac{\ln\left(\frac{K(t)}{1.5}\right)+0.095(6-t)}{0.1\sqrt{6-t}}\right)$$

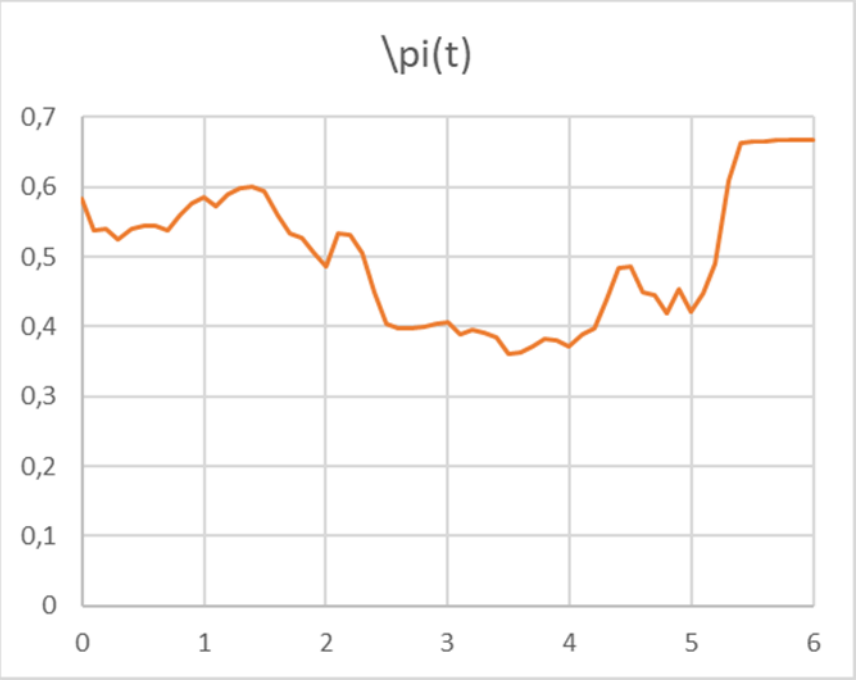
Optimal portfolios with climate scenarios – A path of a simplied example

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=> π^* = 1/3 => R* = 0.2

Hypothetical paths





Optimal portfolios with climate scenarios – Still a lot to do

Remarks:

a) The above example is oversimplified with regard to the evolution of the global temperature. There is need for working with meterologists and other climate scientists.

b) Heinke's rating approach punishes non-sustainable assets only by assigning them a rating of zero. An extension to „strictly negative" rated assets is desirable.

c) In a setting indicated in b), you can also achieve a good sustainability rating by short selling brown assets. As this is undesirable, abandoning short sales of brown assets is considered in Nurkanoviv, K. (2026).

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